Machine Learning (2026) 115:251
https://doi.org/10.1007/s10994-026-07164-1

Weighted UnLearning from Experience for Improved Spurious Correlation Robustness

Jeff Mitchell[1] · **Jesus Martinez-del-Rincon**[1] · **Niall McLaughlin**[1]

Received: 11 March 2026 / Revised: 28 July 2026 / Accepted: 2 September 2026
© The Author(s) 2026

Abstract
Despite their success, Deep Learning (DL) models are susceptible to spurious correlations. These are accidental correlations between non-predictive features of the training dataset and the target label, leading to models that do not generalise well to samples without such spurious correlations. We propose to address this failure mode by extending and improving an existing method for spurious correlation robustness — UnLearning from Experience (ULE). ULE trains two models in parallel; a biased model is encouraged to learn the spurious correlations, while a separate debiased model is trained to solve the same task while avoiding learning the same behaviour as the first, thus avoiding learning the spurious correlations. A weakness of ULE is that it treats every training example identically, regardless of whether or not it contains a spurious correlation. Therefore, ULE does not make optimal use of the training data, leading to worse-than-necessary performance. To address this weakness, we introduce Weighted UnLearning from Experience (wULE), which modifies the original ULE loss function to reweight the components depending on whether a training sample likely contains spurious correlations. Hence, we explore two methods for identifying training samples likely to contain spurious correlations. We test our approach using several well-known spurious correlation robustness benchmarks — Waterbirds, CelebA, UrbanCars and MultiNLI. Our novel wULE method outperforms the original ULE in all cases and achieves Worst-Group Accuracy (WGA) results comparable to the best methods in the spurious correlation robustness literature.

Keywords Spurious correlations · Learning algorithms · Simplicity bias

Editor: Bo Han.

✉ Jeff Mitchell
jmitchell25@qub.ac.uk

Jesus Martinez-del-Rincon
j.martinez-del-rincon@qub.ac.uk

Niall McLaughlin
n.mclaughlin@qub.ac.uk

[1] Centre for Secure Information Technologies (CSIT), Queen's University Belfast, Belfast, United Kingdom of Great Britain and Northern Ireland

Published online: 20 October 2026

1 Introduction

Deep neural networks are typically trained by minimising the average loss over the training set. This approach is commonly known as Empirical Risk Minimization (ERM). Furthermore, the training and testing sets are assumed to be drawn from the same distribution. In practice, this assumption is often violated, meaning the model will encounter Out-of-Distribution (OOD) data. Spurious correlations are a specific type of OOD data and can be seen as a sub-problem related to domain shift (Sagawa et al., 2020; Creager et al., 2021). Spurious correlations occur when the training set contains an accidental correlation between non-predictive features and the target label. For example, if all the images of a certain bird species in the training dataset of a bird species image classifier show the bird on a water background, the classifier, trained using standard training methods, *i.e.*, ERM may learn to predict that species every time an image with a water background is observed. When testing images of the same bird species with different backgrounds are shown to the classifier, its performance will likely drop (Sagawa et al., 2020). This occurs because the classifier has learned the coincidental correlation between the species label and the water background rather than learning the features related to the bird species' appearance. From the ERM-trained classifier's point of view, it cannot tell the difference between truly task-relevant predictive features that will generalise to new situations and features spuriously correlated with the class-label in its training set.

In the general case, it is known that without human supervision, prior information, or assumptions, a general learning algorithm cannot distinguish between truly predictive features and coincidental correlations observed during training (Darbinyan et al., 2023; Izmailov et al., 2022). To tackle this problem, the simplicity bias principle has been commonly invoked in the prior literature. The assumption is that spurious correlations are easier to learn than features related to the primary task (Shah et al., 2020; Yang et al., 2024; Tiwari & Shenoy, 2023). In support of this principle, it has been experimentally demonstrated that neural networks learn features in simple to complex order during training (Tiwari & Shenoy, 2023; Yang et al., 2024), thus showing that neural networks provably exhibit simplicity bias (Shah et al., 2020).

Based on the simplicity bias principle, training paradigms such as the UnLearning from Experience (ULE) algorithm (Mitchell et al., 2025) can address spurious correlations in realistic datasets. ULE trains two models in parallel; the first model uses standard neural network training, *i.e.*, ERM. Under the simplicity bias principle, the ERM-trained model will tend to solve the learning task by taking advantage of spurious correlations in the training set. This trend is especially strong early in training. A second model is then trained to solve the same task while avoiding the behaviour of the first model via a gradient-based loss function. A customised loss function ensures the gradient with respect to the input of the second network differs from the first one. Thus, it does not learn the same spurious correlations. While ULE has been demonstrated to mitigate spurious correlations in realistic datasets successfully, a weakness of the ULE loss function is that it treats every training example in the same way, regardless of whether it is likely to contain a spurious correlation, resulting in frequent violations of the simplicity bias principle. We address this weakness by proposing an enhanced training paradigm and loss function so that different training examples are treated differently depending on whether they are likely to contain spurious correlations. Our contributions are as follows:

- We propose Weighted UnLearning from Experience (wULE), a novel training paradigm which is resilient to spurious correlations, which extends and improves ULE by modifying its loss function to reweight both the classification and gradient components of the loss depending on whether a training example likely contains spurious correlations.
- Our proposed wULE formulation is a generalised form of ULE, which is a particular case of our paradigm.
- We propose a mechanism to estimate the likelihood of individual training samples exhibiting spurious correlations.
- Based on the simplicity bias hypothesis, we explore and compare two methods for estimating the set of samples in the training set likely to contain spurious correlations.
- We propose and evaluate a solution for mitigating the presence of multiple simultaneous spurious correlations.
- We thoroughly evaluate, both quantitatively and qualitatively, the performance of wULE on widely used datasets in the spurious correlation literature, including Waterbirds, CelebA, UrbanCars and MultiNLI datasets. We show that its performance exceeds ULE in all cases and is comparable with the SOTA in the prior literature.

2 Related Work

We classify methods for spurious correlation robustness into single-model (Vapnik, 1991; Sagawa et al., 2020) and multi-model (Nam et al., 2020; Liu et al., 2021; Creager et al., 2021; Sohoni et al., 2020; Ahmed et al., 2021; Kirichenko et al., 2023; Yang et al., 2024; Zhang et al., 2022b) approaches. The latter frequently employ an ERM-trained model for reference or as a bias-amplifying method. Within these two high-level categories, methods can be subdivided into two main types: supervised methods that require group-labels as an essential part of their training process (Sagawa et al., 2020; Kirichenko et al., 2023) and unsupervised methods that do not require group-labels (Vapnik, 1991; Nam et al., 2020; Liu et al., 2021; Creager et al., 2021; Sohoni et al., 2020; Ahmed et al., 2021; Yang et al., 2024; Zhang et al., 2022b). We note that all methods require group-labels for evaluation, and many unsupervised methods tune their hyperparameters using a group-labeled validation set to achieve optimal worst-group accuracy (Liu et al., 2021; Nam et al., 2020; Yang et al., 2024).

Single-model approaches comprise correlation-resilient extensions to the traditional ERM (Vapnik, 1991) method. These approaches (Sagawa et al., 2020; Arjovsky et al., 2019), introduce novel objectives for robust learning, applicable to domain adaptation and generalisation. Group Distributionally Robust Optimization (GroupDRO) (Sagawa et al., 2020) employs supervised group-labels to partition known classification labels into subgroups during training. This partitioning is designed to minimise the worst-case loss within each group by utilising Distributionally Robust Optimization (DRO) for each group, as recommended by Duchi et al. (2021). A main drawback, while ERM only utilises class-labels for optimisation, GroupDRO and DRO both require group-labels in their training objectives, which is unrealistic in real-world settings.

The most successful unsupervised proposed methods opt for a multi-model approach, typically leveraging an ERM-trained model as a reference. Using a reference provides multiple ways to separate training samples into distinct groups, such as estimating correlated and non-correlated samples using either group-labels or correct and incorrectly classified

samples using class-labels. Such methods include: JTT (Liu et al., 2021), LfF (Nam et al., 2020), DFR (Kirichenko et al., 2023), and ULE (Mitchell et al., 2025), which utilise trained ERM models to identify difficult samples in the training set and then train a new model with new learning objectives. JTT and LfF use hard and soft up-weighting of challenging samples, respectively, while LfF uses a generalised cross-entropy loss (Zhang et al., 2018) to amplify the bias of the reference model. DFR (Kirichenko et al., 2023) further determines that retraining only the last layer of the reference model with group-balanced data is adequate to reweight the previously trained features.

Cluster analysis of the logits derived from the ERM reference model is a widely used method that offers distinct class or group-label clusters through silhouette analysis and KMeans (Sohoni et al., 2020; Yang et al., 2024). GEORGE (Sohoni et al., 2020) develops a new model using GroupDRO learning objectives based on these clusters. On the contrary, SPARE (Yang et al., 2024) utilises importance sampling proportional to the sizes of the clustered groups to train a new model. Similarly, CnC (Zhang et al., 2022b) and PruSC (Le et al., 2024) employ a contrastive learning objective to align representations of correlated vs. non-correlated groups and similar spurious attributes. Recent methods include D3M (Jain et al., 2024) which utilise TRAK (Park et al., 2023) to isolate and remove samples that have a negative effect on WGA performance, followed by retraining the model on the remaining samples. EVaLS (Ghaznavi et al., 2024) takes a similar approach by generating a balanced dataset of high and low-loss samples from ERM.

Rather than relying on group-labels, inferred environments, or class-labels to identify specific samples, the Invariant Risk Minimization (IRM) (Arjovsky et al., 2019) and likewise, Pareto IRM (Chen et al., 2023b) principle is applied to the reference model to partition the data into distinct environments or distributions. The partitioned data is then utilised to train a new model using the GroupDRO learning objective or by minimising the Kullback-Leibler (KL) divergence of the partition distributions over each class, as demonstrated by the EIIL (Creager et al., 2021) and PGI (Ahmed et al., 2021) methods, respectively. On the other hand, ULE (Mitchell et al., 2025) simultaneously trains a robust model alongside the reference model, aiming to maximise the distance between the gradients of the reference and robust models.

Additionally, Chen et al. (2023a) identified feature learning itself as a bottleneck for OOD generalisation, while Zhang et al. (2022a) demonstrated that constructing richer representations can mitigate the common trade-off between optimisation and generalisation. These approaches suggest that improving robustness requires not only identifying spurious correlations, but also encouraging the learning of more informative feature representations.

Beyond the literature on spurious correlations, recent work in LLM-based machine unlearning has further highlighted the importance of selectively weighting training samples according to their contribution to undesirable model behaviours. Specifically, Wang et al. (2025) analyse machine unlearning objectives from a gradient perspective and show that different training examples contribute unequally to the forgetting process, motivating gradient-aware optimisation strategies. Similarly, Yang et al. (2025) demonstrate that adaptive loss reweighting based on sample importance can significantly improve unlearning performance.

Despite the multiple advantages of ULE, including the fact it is unsupervised and that its gradient manipulation approach is orthogonal to existing methods, it does not split its training samples into estimated groups. It, therefore, treats all training examples identically,

whether or not they likely contain a spurious correlation. This can lead to worse performance, as the model does not optimally learn from the training data. We, therefore, propose to update the ULE loss function to merge the advantages of multi-model and unsupervised approaches, particularly (Nam et al., 2020; Yang et al., 2024). This will allow ULE to change its loss per sample depending on the likelihood of the presence of a spurious correlation, improving its performance.

3 Background

We first provide an overview of UnLearning from Experience (ULE) (Mitchell et al., 2025), which serves as the foundation for our proposed wULE method. In ULE, two networks, a biased network, $b(x)$, and a debiased network, $d(x)$, are trained together to solve a classification task such that the debiased network gains robustness to spurious correlations in the training dataset. Both networks are trained in parallel and always observe the same training batches.

For a classification task, given a set of training examples, $\{(x_1, \hat{y}_1) \dots (x_i, \hat{y}_i) \dots (x_I, \hat{y}_I)\}$, the goal is to predict label $\hat{y}_i$, given input x_i. In ULE, the biased network is trained to perform classification by minimising the average loss over the training set, *i.e.*, standard neural network training (ERM). Summing over all indices, $i \in I$, in the training set, the biased network's loss function can be written as follows:

$$\mathcal{L}_b = \sum_i \text{CE}(\hat{y}_i, y_i^b) \tag{1}$$

where CE is the cross-entropy loss function, $y_i^b = b(x_i)$ is the biased network's prediction given x_i, and $\hat{y}_i$ is the corresponding target label. The underlying assumption of ULE is that the biased network, trained using ERM, will inevitably exploit spurious correlations in the dataset.

In parallel, the debiased network is trained using a loss function that encourages it to solve the classification task while using different features from the biased network, thus avoiding learning the spurious correlations of the biased model. The features used by the biased network are identified by computing $\frac{\partial b(x)}{\partial x}$, the gradient of $b(x)$ with respect to the input, x, which gives a feature saliency map, g^b, for the biased network. The corresponding saliency map, g^d, for the debiased network can be computed similarly. By inverting the biased network's saliency map, g^b, the Mean-Squared Error (MSE) loss function can be used to encourage the debiased network's saliency map to oppose that of the biased network:

$$\begin{aligned} g^b &= \frac{\partial b(x)}{\partial x} \\ g^d &= \frac{\partial d(x)}{\partial x} \\ \text{MSE}(-g^b, g^d) &= \| - g^b - g^d \|_2^2 \end{aligned} \tag{2}$$

The final ULE loss function for the debiased network, summing over all the indices, i, in the training set, can be written as follows:

$$\mathcal{L}_d = \sum_i \lambda \text{CE}(\hat{y}_i, y_i^d) + (1-\lambda)\text{MSE}(-g_i^b, g_i^d) \tag{3}$$

where $y_i^d = d(x_i)$ is the debiased network's output given x_i. A hyperparameter λ controls the relative weighting of both components of the loss function.

The core assumption of ULE is the simplicity bias hypothesis, *i.e.*, the spurious correlations are easier to learn than the primary task. The simplicity bias assumption has been experimentally demonstrated in the prior literature (Tiwari & Shenoy, 2023; Yang et al., 2024). And it has been theoretically shown that neural networks provably exhibit simplicity bias (Shah et al., 2020). Hence, in ULE, the biased network, $b(x)$, trained using ERM, Equation 1, is expected to focus on learning the spurious correlations, especially early in training. Therefore, the ULE loss function, Equation 3, applied to the debiased network, $d(x)$, encourages it to learn different features from the biased network while still solving the same primary task, thus causing $d(x)$ to gain robustness by avoiding the spurious correlations of the biased network.

4 Weighted UnLearning from Experience (wULE)

An issue for ULE is that the loss function, Equation 3, is the same for all training examples. Thus, it implicitly assumes that spurious correlations are present in all training examples. In practice, this assumption is often violated, as real datasets will contain examples with and without spurious correlations. Therefore, the ULE loss function, which encourages the debiased network to diverge from the biased network in all cases, will hinder learning on examples without spurious correlations by forcing the predictions of both networks apart when they should remain identical.

To address this issue, we first assume there exists an index set, $\mathcal{S}$, the spurious correlation set, containing indices, $i \in \mathcal{S}$, of all examples, x_i, with spurious correlations in the training set. We can then rewrite Equation 3, the loss function for the debiased network, in terms of indices $i \in \mathcal{S}$ and all remaining indices, $i \notin \mathcal{S}$:

$$\begin{aligned}\mathcal{L}_d = \lambda\big(\alpha \sum_{i \notin S} \text{CE}(\hat{y}_i, y_i^d) + \sum_{i \in S} \text{CE}(\hat{y}_i, y_i^d)\big) + \\ (1-\lambda)\big(\beta \sum_{i \in S} \text{MSE}(-g_i^b, g_i^d) + \sum_{i \notin S} \text{MSE}(-g_i^b, g_i^d)\big)\end{aligned} \tag{4}$$

Equation 4 is a generalised form of the ULE loss function with scaling factors α and β applied depending on whether a given training example, x_i, contains a spurious correlation. We restrict the hyperparameters α and β to positive values only, as negative weights would effectively maximise, rather than minimise, the loss, undermining the learning objective.

We name this generalised loss function *Weighted UnLearning from Experience (wULE)*. In contrast to ULE, where the loss encourages the biased and debiased networks to always behave differently, the goal of the wULE loss is to train the debiased network to perform the classification task while ensuring behaviour differs from the biased network only on samples likely to contain spurious correlations. Figure 1 illustrates our proposed approach.

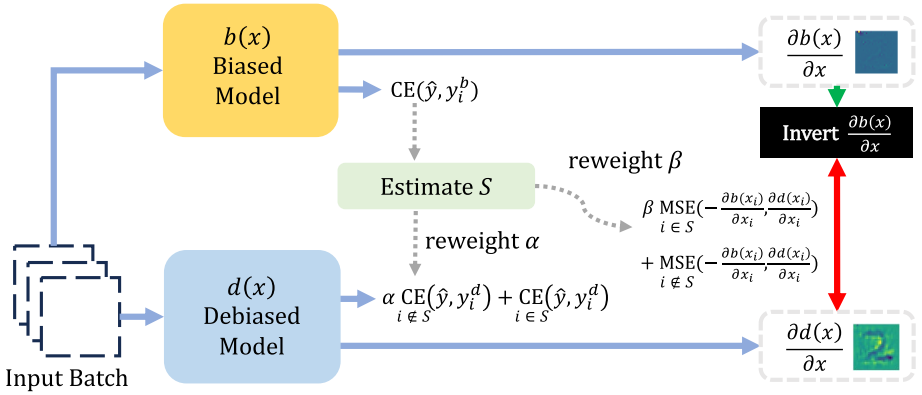


Fig. 1 Weighted UnLearning from Experience (wULE). Two models are trained in parallel: a biased model, $b(x)$, and a debiased model, $d(x)$. The ultimate goal is for $d(x)$ to be robust to spurious correlations. To achieve this, $b(x)$ is trained via ERM, and hence, via the simplicity bias assumption, it will tend to solve the classification task using spurious correlations. Then, $d(x)$ is trained to solve the same classification task while avoiding the mistakes as of $b(x)$ by observing $\frac{\partial b(x)}{\partial x}$ and ensuring its own gradient $\frac{\partial d(x)}{\partial x}$ is the opposite. The classification and gradient components of $d(x)$'s loss function are reweighted per example, depending on an estimate of, $\mathcal{S}$, the set of training examples containing spurious correlations. Hence, the wULE loss is adjusted per training example depending on whether it likely contains spurious correlations. The final trained $d(x)$ network will then have increased spurious correlation robustness

For the classification components of the wULE loss in Equation 4, training examples without a spurious correlation, *i.e.*, x_i where $i \notin S$, can be regarded as more challenging to classify correctly due to the simplicity bias hypothesis (See Sections 3 and 4.1). Therefore, the scaling factor α allows the network to place greater relative emphasis on the challenging training examples without spurious correlations versus easier examples with spurious correlations.

For the feature saliency components of the wULE loss in Equation 4, training examples with spurious correlations, *i.e.*, x_i where $i \in S$, should be treated differently. For these examples, greater emphasis should be placed on encouraging the debiased network to avoid the spurious correlations present in the biased network. This can be achieved by scaling β to increase the weighting of the loss components for examples with spurious correlations relative to examples without.

Finally, the λ hyperparameter controls the relative weighting of the classification and feature-saliency components of the wULE loss function, regardless of the content of the training examples.

4.1 Estimation of Spurious Correlation Set

So far, we have assumed that $\mathcal{S}$, the set of indices of training examples with spurious correlation, is known. In practical settings, ground-truth group-label annotations will not be available. We therefore explore two ways of estimating $\mathcal{S}$.

4.1.1 JTT-Based

Under this approach, we populate $\mathcal{S}$ with the indexes of all training examples correctly classified by the biased network, $b(x)$, early in training, *i.e.*, at epoch k. As the biased network, $b(x)$, is trained using ERM, under the simplicity bias hypothesis (See Section 3), it will tend to solve the classification task by focusing on spuriously correlated features early in its training process. Hence, samples with spurious correlations will be easier for the biased network to classify, meaning it will tend to classify these examples correctly. We, therefore, populate $\mathcal{S}$ with all samples from the training set correctly classified by the biased network at epoch k. This approach is equivalent to the error-set estimation method used in JTT (Liu et al., 2021). As with JTT, we freeze the biased network after epoch k and do not update its parameters further, while the debiased network continues to learn until the end of training.

4.1.2 SPARE-Based

Alternatively, we populate $\mathcal{S}$ using the group-inference method proposed by SPARE (Yang et al., 2024). This method infers group-labels, *i.e.*, the group of training samples with and the group without spurious correlations, by clustering the logits of the training examples from the ERM-trained biased network at epoch k, where k is early in the training process. The intuition is that training examples that are members of the same group will be represented via similar activation patterns within the network, especially during the early stages of the training process. SPARE is also based on the simplicity bias hypothesis (See Section 3). It assumes that early in the training process, the network's internal representations will depend more on spurious correlations than relevant features. Hence, the logit values of all training examples can be clustered to reveal different groups of training examples. Following the procedure of SPARE, we assume that the majority group represents samples with spurious correlations. We freeze the biased network after epoch k, while the debiased network continues to learn until the end of training.

4.2 Practical Implementation

In common with other methods in the spurious correlation literature, such as JTT (Liu et al., 2021), EIIL (Creager et al., 2021), SPARE (Yang et al., 2024), ULE (Mitchell et al., 2025) *etc.*, we assume that a pre-trained image classification model is available (Joshi et al., 2023). It has been shown in the spurious correlation literature that fine-tuning the final layer representation of a pre-trained model is sufficient for spurious correlation robustness (Kirichenko et al., 2023). Hence, both our biased and debiased networks are initialised using the weights of a model pre-trained on a general image classification dataset such as COCO (Lin et al., 2014) or ImageNet (Deng et al., 2009). During wULE training, we only fine-tune the last layer representations of both networks while the earlier network layers are frozen. After wULE training, the biased model, $b(x)$, is discarded. Only the debiased model, $d(x)$, trained via wULE to be robust to spurious correlations, is used for inference. Hence, while wULE requires training two models in parallel, this is only required for the first k epochs. Furthermore, at inference time, only one model is required, so, there is no inference time computational cost penalty.

We also note that unlike JTT (Liu et al., 2021), and SPARE (Yang et al., 2024), the networks $b(x)$ and $d(x)$ are trained concurrently for the first k epochs. Then the weights of $b(x)$ are frozen for the remainder of training process while the weights of $d(x)$ continue to update until the end of training.

5 Experiments

This section details the evaluation and fine-tuning of our proposed wULE approaches. In common with related approaches in the literature, we use the ResNet-50 and ResNet-18 (He et al., 2016) pre-trained models for our experiments. We perform spurious correlation robustness experiments on datasets including Waterbirds (Wah et al., 2011; Zhou et al., 2018), CelebA (Liu et al., 2015) and the newly introduced UrbanCars dataset (Li et al., 2023) which features two different types of concurrent spurious correlations. Additionally, we leverage a BERT model, specifically the Hugging Face implementation of the bert-base-uncased model (Devlin et al., 2019; Wolf et al., 2020) for the Natural Language Inference (NLI) dataset — MultiNLI (Williams et al., 2018).

5.1 Datasets

In our experiments, we use the following datasets common in the spurious correlation literature. For all datasets, we use the default splits for training, validation and testing provided by Sagawa et al. (2020); Li et al. (2023).

5.1.1 Waterbirds

This dataset shows a correlation between image backgrounds and image foregrounds. The dataset features images of birds where the classification labels $y = \{\text{waterbird}, \text{landbird}\}$, are spuriously correlated with the image background $a = \{\text{water background}, \text{land background}\}$. The majority of samples show landbirds on land backgrounds and waterbirds on water backgrounds, *i.e.*, with spurious correlations. Only around five percent of the dataset images show landbirds on water backgrounds and waterbirds on land backgrounds, *i.e.*, no spurious correlations.

Training a classification model on Waterbirds using ERM will tend to result in a model that associates the label *waterbird* with the water background and the label *landbird* with the land background. This behaviour of the ERM algorithm minimises the average training loss across the training set but leads to poor performance on the non-spuriously correlated groups in the dataset.

5.1.2 CelebA

The CelebA dataset contains a different type of spurious correlation. The images contain celebrity faces that should be classified as having a specific hair type $y = \{\text{blond hair}, \text{non-blond hair}\}$ with the spurious correlation $a = \{\text{female}, \text{male}\}$ respectively. Most of the images in the dataset show females with blonde hair and males

with non-blond hair. Approximately one percent of the dataset consists of images free from spurious correlations, *i.e.*, males with blond hair.

5.1.3 UrbanCars

The recently introduced UrbanCars dataset contains two spurious correlations. The primary task is to classify images of cars (foreground) into the labels $y = \{\text{urban}, \text{country}\}$. The images also contain a spurious correlation with the background, which can have values $a_1 = \{\text{urban}, \text{country}\}$, and finally, a co-occurring object placed into the images, which can also take values of $a_2 = \{\text{urban}, \text{country}\}$. The majority groups are samples where $(y, a_1, a_2) = (\text{urban}, \text{urban}, \text{urban})$ and $(y, a_1, a_2) = (\text{country}, \text{country}, \text{country})$. The non-correlated groups comprise the other six combinations of (y, a_1, a_2). These group combinations contain samples with varying severity of spuriously correlated features, *i.e.*, only one of $a_1 == y$ or $a_2 == y$ is present, followed by the most severe combination where neither $a_1 == y$ nor $a_2 == y$ is present.

5.1.4 MultiNLI

The MultiNLI dataset is a Natural Language Inference benchmark where the objective is to determine the relationship between two sentences, a premise and a hypothesis, where the relationship is labelled as one of $y = \{\text{entailment}, \text{neutral}, \text{contradiction}\}$. Following (Sagawa et al., 2020; Liu et al., 2021), we consider the presence of negation words (*e.g.*, no, not, never, nothing) in the hypothesis as a spurious attribute $a = \{\text{no negation}, \text{negation}\}$, as due to annotation artifacts introduced during dataset collection, contradiction examples are disproportionately associated with negation terms. The addition of this text-based dataset allows for evaluation of wULE's ability to generalise to domains other than images.

5.2 Evaluation Metrics

Following convention, we evaluate spurious correlation robustness using WGA (Sagawa et al., 2020) on the testing set. We assume group-labels are only available for evaluation purposes. To calculate WGA, the validation set group-labels are used to calculate the classifier's accuracy within each group. The WGA is then the classifier's accuracy on the worst-performing group.

A robust classifier is expected to be unaffected by the presence or absence of spurious correlations, *i.e.*, it will be unaffected by the group, resulting in high WGA, similar to all other sample groups. On the contrary, a non-robust model will only perform well on groups containing spurious correlations, leading to a diverge of performance for all other groups and thus reducing WGA. Hence, WGA encourages optimisation for worse case performance, *i.e.*, performance on the group with no spurious correlations. We also report average accuracy as a reference, however, it is arguable that average accuracy can be misleading, whereas WGA better reflects a model's susceptibility to spurious correlations. Le et al. argue *"A significant drop between average accuracy and worst-group accuracy is a strong indicator of relying on spurious correlations"* (Le et al., 2024) leading to worse performance on subgroups where the spurious correlations don't hold. Hence, we optimise for high WGA and a similar average accuracy.

During our evaluation, we only use the group-labels in the validation set for early stopping and hyperparameter search. This is the approach commonly taken in the literature (Nam et al., 2020; Liu et al., 2021; Creager et al., 2021; Sohoni et al., 2020; Ahmed et al., 2021; Yang et al., 2024; Zhang et al., 2022b) even for unsupervised methods. We also note that some other methods rely on group-labels during training and validation as a fundamental element of their supervised approach (Kirichenko et al., 2023; Sagawa et al., 2020). We argue that such methods are more limited than ours as group-labels will not be available in all circumstances, especially in domains with multiple and/or unknown spurious correlations, which are difficult or impossible to annotate unambiguously.

5.3 Hyperparameter Tuning

We use the ResNet-50 model pre-trained on the "ImageNet-1K" dataset as the backbone for our biased and debiased models. The biased model, $b(x)$, was trained using ERM, while the debiased model, $d(x)$, used the wULE loss (See Section 4). In all cases, we train only the final fully-connected layer of the model. This final layer fine-tuning approach has been demonstrated to be sufficient for achieving spurious correlation robusntess (Kirichenko et al., 2023).

As with related approaches in the literature (Liu et al., 2021; Sagawa et al., 2020; Yang et al., 2024), the best hyperparameters were tuned independently for each dataset via WGA on the validation set with grid search. For image-based datasets, learning rates were searched over the range $[1e-1, 1e-2, 1e-3, 1e-4]$ and weight decay values over $[0, 1e-1, 1e-2, 1e-3, 1e-4]$. While, for the text-based MultiNLI dataset and BERT model, a linearly decreasing learning rate initialised at $2e-5$ with no weight decay was used. For the remaining hyperparameters, λ was searched over the range [0.001, 0.01, 0.1, 0.3, 0.5, 0.7, 0.9, 0.99, 0.999], and scaling factors α and β over the range [0, 1, 2, 5, 10, 20, 50, 100] for all datasets. Additionally, models were trained for 300 and 5 epochs with a batch size of 128 using the Adam (Kingma & Ba, 2015) and AdamW (Loshchilov & Hutter, 2019) optimisers for the image and text-based datasets respectively. Models were implemented using PyTorch (Paszke et al., 2019) running on NVIDIA Tesla V100 GPUs. The best hyperparameters for each dataset are shown in Table 1.

5.4 Oracle Model

In this experiment, we measure the theoretical maximum performance of wULE, assuming that ground-truth group-labels are available, *i.e.*, we assume to know whether or not each training example contains a spurious correlation. There is, therefore, no error in the estimation of $\mathcal{S}$. We compare the theoretical maximum performance with the actual performance, where group-labels are estimated based on either the simplicity bias assumption (JTT-Based) or group-inference via clustering techniques (SPARE-Based) (See Section 4.1).

We call the model that uses the ground-truth group information the Oracle model. In the Oracle model, we use ground-truth information to populate $\mathcal{S}$ with the indexes of all training examples that contain spurious correlations. We compare the performance of the Oracle model with two variants of wULE, where $\mathcal{S}$ is estimated using the methods outlined in Section 4.1. We also show results obtained by standard ULE as a baseline. In all cases, we use

Table 1 Best hyperparameters for the Waterbirds, CelebA, UrbanCars and MultiNLI datasets. For each dataset, we optimised either α or β independently, or both α and β together

	Waterbirds			CelebA			UrbanCars			MultiNLI		
Hyperparameter	**Oracle**	**SPARE**	**JTT**	**Oracle**	**SPARE**	**JTT**	**Oracle**	**SPARE**	**JTT**	**Oracle**	**SPARE**	**JTT**
Freeze Epoch k	0	9	8	0	12	12	0	4	7	0	3	3
Learning Rate	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.00002	0.00002	0.00002
Weight Decay	0.001	0.1	0.1	0.01	0.01	0.01	0.0001	0.1	0.1	0	0	0
ULE λ	0.9	0.4	0.4	0.999	0.1	0.1	0.1	0.99	0.99	0.5	0.1	0.1
Scaling Factor α	100	20	20	20	50	100	1	100	50	50	20	5
Scaling Factor β	5	50	100	5	100	20	50	50	5	5	10	5

hyperparameter search over the validation set to optimise the performance of each method. We then report results on the testing set. The results of this experiment are shown in Table 2.

When comparing wULE with different spurious correlation set estimations to the Oracle model in Table 2, it becomes evident that the Oracle model establishes the theoretical maximum performance that can be achieved by wULE. The actual wULE results, using either estimation variant, for WGA significantly improve over ULE, and come close to the Oracle model. This shows the effectiveness of the wULE algorithm, which does not require prior knowledge of the spurious correlations and their group-labels to achieve performance comparable to the Oracle. It further emphasises the effectiveness of our proposed methods for estimating $\mathcal{S}$ (this is further analysed in Section 5.9, both of which perform comparably, with the SPARE-based method, performing slightly better. Overall, these results show that through the additional new terms in the loss for weighting based on the presence of spurious correlations, the performance of ULE can be significantly improved.

5.5 Estimation of Spurious Correlation Set

As outlined in Section 4.1, there are several ways to estimate the spurious correlation set $\mathcal{S}$, which contains the indices, $i \in \mathcal{S}$, of all training samples, x_i, with spurious correlations. Its complement, denoted $\mathcal{S}'$, contains the remaining training examples without spurious correlations. The sets $\mathcal{S}$ and $\mathcal{S}'$ are mutually exclusive and collectively exhaustive, such that $\mathcal{S} \cap \mathcal{S}' = \emptyset$ and $\mathcal{S} \cup \mathcal{S}'$ comprises all training samples. In this section, we investigate the accuracy of the different estimation methods for $\mathcal{S}$.

Using the Waterbirds dataset, we train the biased network, $b(x)$, via ERM for k epochs and then freeze it. Given the frozen biased network, we estimate $\mathcal{S}$ using either the JTT (Liu et al., 2021) or SPARE-based methods (Yang et al., 2024), outlined in Section 4.1. Until then, $\mathcal{S}$ is a continuously changing set of indices before the desired k epochs are achieved. We then continue training the debiased network, $d(x)$, to convergence using the estimated $\mathcal{S}$, which remains fixed. We repeat this procedure for all values of k and report the final WGA of the debiased model. Results were averaged over five runs with different random seeds and are reported in Figure 2 (bottom).

In Figure 2 (top), for every value of k, we measure how well our estimated partitioning of the training data into the mutually exclusive sets $\mathcal{S}$ (spurious) and $\mathcal{S}'$ (non-spurious) matches the corresponding ground-truth. Assigning a training example to either $\mathcal{S}$ or $\mathcal{S}'$ can be viewed as a binary classification task. We, therefore, evaluate the correspondence between the estimated and ground-truth partitions using F1-score, treating $\mathcal{S}'$, the set of samples without spurious correlations, as the positive (rare) class. The use of the F1-score is appropriate given the imbalanced nature of the Waterbirds dataset, which primarily comprises samples in $\mathcal{S}$.

Table 2 Results for the Oracle model compared to the baseline ULE and wULE methods on the Waterbirds dataset using the JTT and SPARE derived method for estimating $\mathcal{S}$

	Worst-Group Accuracy	Average Accuracy
ULE	$86.5_{\pm 0.2}$	$89.8_{\pm 0.2}$
wULE (JTT)	$89.4_{\pm 0.4}$	$90.3_{\pm 0.4}$
wULE (SPARE)	$89.8_{\pm 0.6}$	$90.9_{\pm 0.3}$
wULE (Oracle)	$91.0_{\pm 0.1}$	$92.2_{\pm 0.1}$

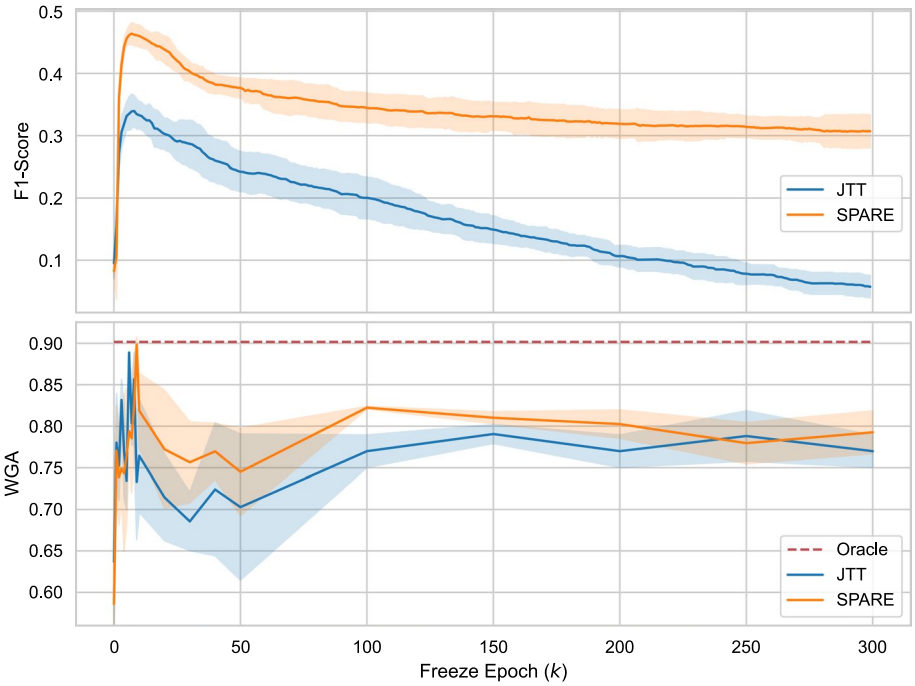


Fig. 2 Top: Agreement, as measured using F1-score, between the estimated partition of the training examples into $\mathcal{S}$ (spurious) and $\mathcal{S}'$ (non-spurious) and the corresponding ground-truth partition. Bottom: Worst-Group Accuracy (WGA) of the debiased model using the corresponding spurious correlation set estimation method. In both cases, top and bottom, results are reported after freezing the biased network at epoch k and training the debiased model to convergence

The results in Figure 2 show that both JTT and SPARE methods for estimating $\mathcal{S}$ achieve peak performance when the biased network is frozen early in the training process. This supports the simplicity bias assumption that the biased network focuses on spurious correlations early in training. The JTT and SPARE methods of estimating $\mathcal{S}$ peak at similar k values, however, the SPARE method consistently outperforms JTT. The JTT method becomes uninformative at later epochs as its ability to estimate $\mathcal{S}$ diminishes. This happens because the biased network eventually overfits to the full training set, so it tries to include all samples in $\mathcal{S}$. On the other hand, the SPARE-based method still retains some usefulness, albeit smaller in later values of the freeze epoch, since the clustering mechanism still forces the estimation of $\mathcal{S}$ to be divided into two distinctive groups.

In terms of WGA, we observe that when the estimate of $\mathcal{S}$ is better, the WGA of the debiased network tends to improve. The peaks of the graphs for estimating $\mathcal{S}$ and WGA occur at the same k value. The SPARE method for estimating $\mathcal{S}$ is preferable, as it leads to better WGA and is less sensitive to choosing the correct k value. Finally, this experiment shows there exists an optimal epoch k at which the biased network should be frozen to best estimate $\mathcal{S}$. This value was reported in Table 1.

Finally, we extend our analysis to the other datasets studied. We report the WGA and average accuracy for the JTT and SPARE methods for estimating the spurious correlation set $\mathcal{S}$ across the Waterbirds, CelebA, UrbanCars and MultiNLI datasets. The results in

Table 3 Worst-Group Accuracy (WGA) and average accuracy for the JTT and SPARE wULE variants for estimating the spurious correlation set $\mathcal{S}$ on the Waterbirds, CelebA, UrbanCars and MultiNLI datasets

	Waterbirds		**CelebA**		**UrbanCars**		**MultiNLI**	
	Worst-Group	Average	**Worst-Group**	Average	**Worst-Group**	Average	**Worst-Group**	Average
wULE (JTT)	$89.4_{\pm 0.4}$	$90.3_{\pm 0.4}$	$87.4_{\pm 0.4}$	$88.5_{\pm 0.2}$	$75.7_{\pm 0.7}$	$96.6_{\pm 0.4}$	$73.8_{\pm 0.6}$	$80.9_{\pm 0.3}$
wULE (SPARE)	$89.8_{\pm 0.6}$	$90.9_{\pm 0.3}$	$88.0_{\pm 0.3}$	$88.4_{\pm 0.3}$	$76.2_{\pm 0.6}$	$96.6_{\pm 0.4}$	$75.0_{\pm 0.5}$	$81.2_{\pm 0.3}$

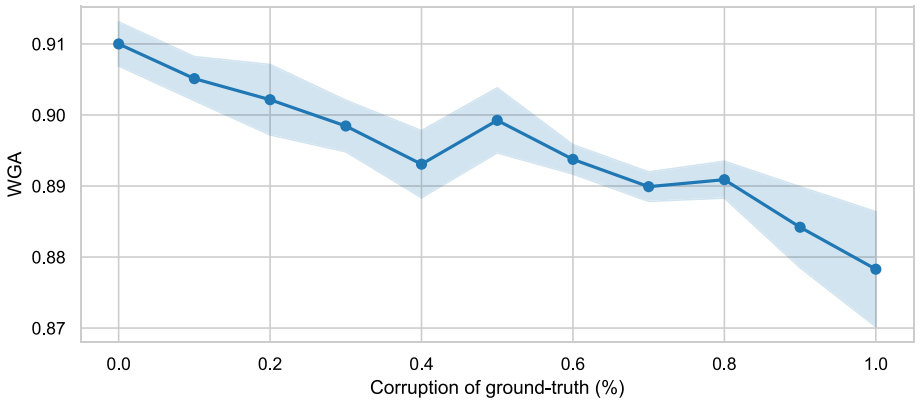


Fig. 3 Sensitivity of wULE to errors in the estimation of $\mathcal{S}$, the set of samples containing spurious correlations. Based on an Oracle experiment, starting with the correct $\mathcal{S}$, *i.e.*, randomly corrupting a percentage of labels. Performance measured using Worst-Group Accuracy (WGA) on Waterbirds

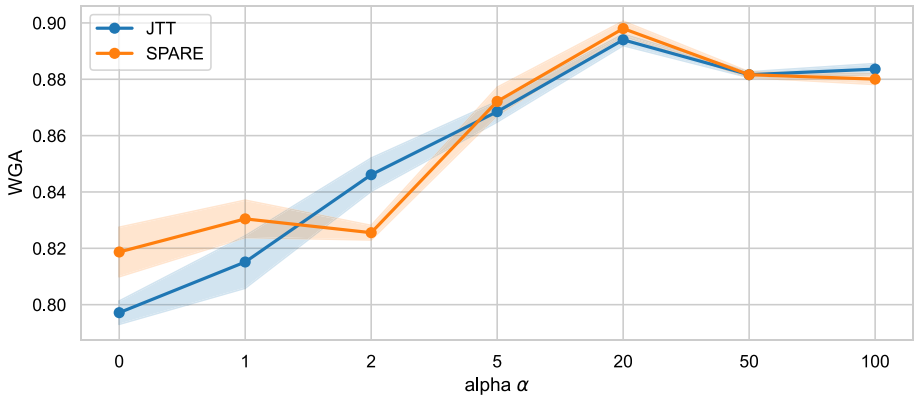


Fig. 4 Sensitivity of hyperparameter α associated with the classification terms of the wULE loss in Equation 4

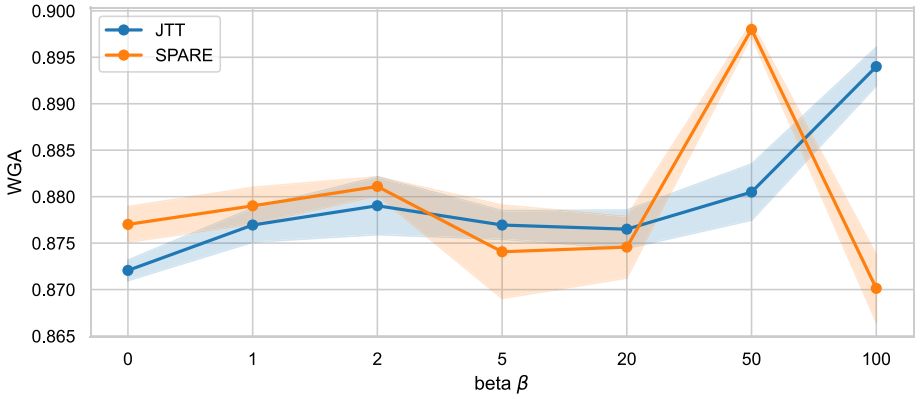


Fig. 5 Sensitivity of hyperparameter β associated with the MSE terms of the wULE loss in Equation 4

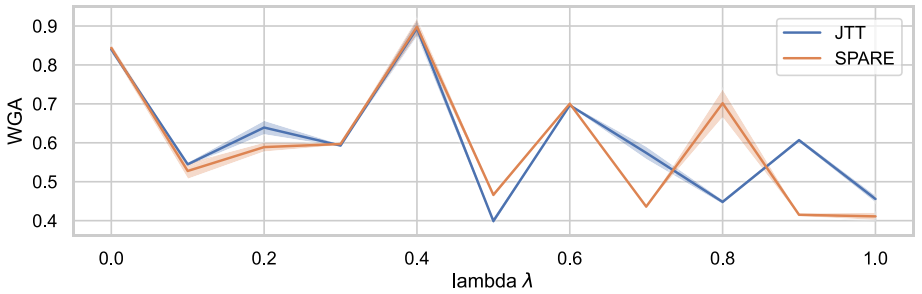


Fig. 6 Sensitivity of the hyperparameter λ associated with the ULE term of the wULE loss in Equation 4

Table 3 show that, in common with its performance on Waterbirds, in terms of WGA, the SPARE method for estimating $\mathcal{S}$ outperforms the JTT-based method across all the datasets. Overall, the results here underscore the importance of accurately estimating $\mathcal{S}$ for wULE to achieve optimal performance.

5.6 Sensitivity to Spurious Correlation Set

In the previous experiment (See Section 5.9) we observed that wULE's peak WGA occurs simultaneously with peak $\mathcal{S}$ set estimation (See Figure 2). However, $\mathcal{S}$ set estimation is often inaccurate for both the SPARE and JTT methods. Therefore, we investigate the sensitivity of wULE to errors in estimating $\mathcal{S}$. Starting with an $\mathcal{S}$ populated using ground-truth information on the samples containing spurious correlations, we progressively corrupt $\mathcal{S}$ by randomly flipping the labels, then measure the WGA on the Waterbirds dataset. In Figure 3, we show how WGA changes as the level of $\mathcal{S}$ set corruption increases on the Waterbirds dataset.

At the maximum corruption level, performance is similar to ULE, while at the minimum corruption level, performance is similar to the Oracle method (See Section 5.8). We also note that at levels of corruption similar to those measured in the previous experiment, *i.e.*, around 60%, the measured WGA is close to the SPARE and JTT results reported in Figure 2 and Table 3.

5.7 Hyperparameter Sensitivity

In the wULE loss in Equation 4 the hyperparameter, α is associated with the cross-entropy terms, and β associated with the MSE terms. An additional hyperparameter λ controls the relative weighting of the classification and wULE feature saliency loss terms. To understand how these parameters influence performance, we take the best hyperparameters found in Section 5.7, and, keeping all other hyperparameters constant, we systematically vary, α and β, one-by-one while training the networks again from scratch using wULE. Each parameter is varied over the range [0, 1, 2, 5, 10, 20, 50, 100], and the final debiased network's WGA is reported. This experiment uses the Waterbirds dataset. The results for α and β are shown in Figures 4 and 5 respectively, and finally, the results for λ are shown in Figure 6.

The hyperparameter α weights the cross-entropy loss between spurious and non-spurious samples. Figure 4 shows that performance increases with larger α values, placing more emphasis on the non-spurious samples. We know that such samples occur infrequently in

Table 4 Comparison between Mean-Squared Error and Mean-Absolute Error loss functions for comparing the saliency maps of the biased and debaised models in Equation 4. We report the Worst-Group Accuracy (WGA) and Average Accuracy for the JTT and SPARE variants of wULE

	Mean-Squared Error (L2)		**Mean-Absolute Error (L1)**	
	Worst-Group	Average	**Worst-Group**	Average
wULE (JTT)	$89.4_{\pm 0.4}$	$90.3_{\pm 0.4}$	$88.7_{\pm 0.3}$	$91.4_{\pm 0.2}$
wULE (SPARE)	$89.8_{\pm 0.6}$	$90.9_{\pm 0.3}$	$89.4_{\pm 0.4}$	$91.9_{\pm 0.3}$

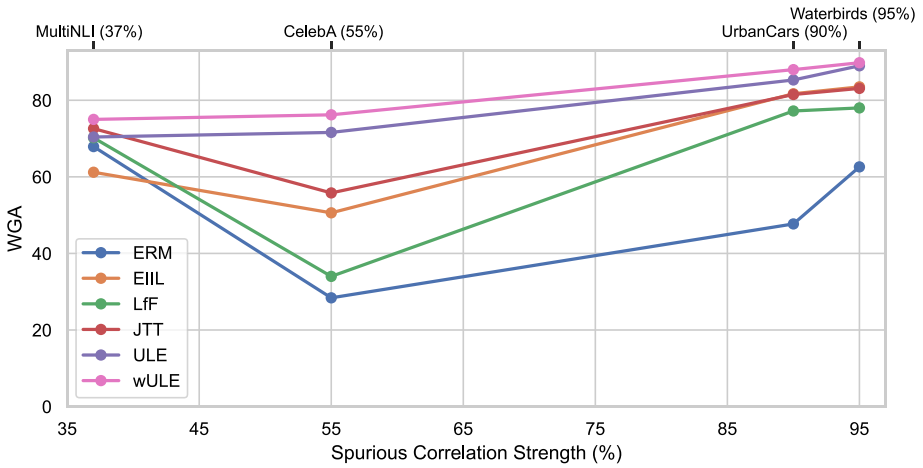


Fig. 7 Worst-Group Accuracy (WGA) of wULE and five other state-of-the-art methods on the Waterbirds, CelebA, UrbanCars and MultiNLI datasets. The spurious correlation strength is calculated as the percentage of training samples that contain spurious correlations

the Waterbirds dataset and are thus more challenging, as the model cannot rely on shortcut learning via spurious features. Hence, placing more emphasis on these hard-to-learn samples makes intuitive sense.

The hyperparameter β weights the MSE loss of samples with and without spurious correlations, respectively. Figure 5 demonstrates that optimal performance occurs when β is 100 and 50 for JTT and SPARE approaches, respectively, aligning with Table 1. Larger β values place increased emphasis on the MSE loss associated with samples containing spurious correlations. This indicates that encouraging the model to push apart the saliency maps of the biased and debiased networks for samples with spurious correlations improves performance. Pushing apart the saliency maps encourages the debiased network to learn different behaviour from the biased network, avoiding learning spurious feature correlations. Hence, the fact that larger values β are associated with improved performance provides evidence that the reasoning behind our proposed wULE loss is correct.

The optimal λ occurs at 0.4, indicating that greater emphasis should be placed on the wULE feature saliency loss term than the classification terms for optimal performance.

Overall, these results support our assertion in Section 4 when proposing wULE that samples with and without spurious correlations should be treated differently. Doing so allows for better WGA and, hence, increased robustness to spurious correlations.

Table 5 Worst-Group Accuracy (WGA) for different combinations of debiased model $\mathbf{d}(\mathbf{x})$ and biased model $\mathbf{b}(\mathbf{x})$ architectures on the Waterbirds dataset using the JTT-based method for estimating $\mathcal{S}$

	$b(x)$		
$d(x)$	**ResNet-18**	**ResNet-50**	**ViT-H-14**
ResNet-18	87.1	87.2	87.5
ResNet-50	90.0	89.4	89.7
ViT-H-14	94.1	94.1	94.4

Table 6 Worst-Group Accuracy (WGA) for different combinations of debiased model $d(x)$ and biased model $b(x)$ architectures on the Waterbirds dataset using the SPARE-based method for estimating $\mathcal{S}$

	$b(x)$		
$d(x)$	**ResNet-18**	**ResNet-50**	**ViT-H-14**
ResNet-18	88.1	87.9	88.7
ResNet-50	91.2	89.8	90.1
ViT-H-14	94.1	94.2	94.7

5.8 Loss Function Investigation

The loss function used by wULE (and ULE) to compare the saliency maps of the biased and debiased models is the L2 norm, *i.e.*, the Mean-Squared Error between the pixels of the saliency maps of the biased and debiased models (See Equation 4). Here we investigate the performance of wULE when using the L1 norm, *i.e.*, Mean-Absolute Error defined as $\text{MAE}(-g^b, g^d) = \| - g^b - g^d \|_1$. The L1 norm is known to promote sparse solutions, which may help the model to focus only on the most relevant parts of the saliency maps. We report our results, comparing both approaches, on the Waterbirds dataset in Table 4.

The results in Table 4 show that in terms of WGA, while MAE's performance is marginally below MSE, using MAE slightly improves average accuracy. In this work, we focus on WGA, so we use MSE for all further experiments.

5.9 Spurious Correlation Strength

In this section, we investigate the effect of spurious correlation strength on the performance of wULE. For the four datasets from Section 5.1, we measure the strength of spurious correlations in each dataset and plot it against performance measured by WGA. We quantify the strength of spurious correlations as the percentage of training samples that contain spuriously correlated features. The strengths vary from MultiNLI (37%), CelebA (50%), UrbanCars (90%) to Waterbirds (95%). Figure 7 shows the WGA achieved by wULE and five other state-of-the-art methods across all four datasets.

In Figure 7, we see that wULE consistently performs well across all datasets, outperforming the other methods regardless of spurious correlation strength.

5.10 Mixed Models

Our wULE approach trains two models in parallel, and so far we have assumed that the models share the same architecture. We now investigate how wULE performs with different model architectures. We explore combinations of ResNet-18, ResNet-50 and ViT-H-14 for

the biased and debiased model architectures. We report the results on the Waterbirds dataset in Tables 5 and 6 for the JTT and SPARE approaches of estimating $\mathcal{S}$, respectively.

The results in Tables 5 and 6 show that the SPARE-based method for estimating the spurious correlation set, $\mathcal{S}$, consistently performs better than the JTT-based approach. Overall, the best performance occurs when the ViT-H-14 model is used for both the biased and debiased models. The trend along the diagonal of Tables 5 and 6 shows that performance decreases as simpler models, ResNet-50 and ResNet-18, are used.

Reading horizontally across Tables 5 and 6 shows that for a fixed complexity of debiased model, $d(x)$, increasing the complexity of the biased model, $b(x)$, does not significantly impact performance since its role is to learn misleading features and capture and promote spurious correlations. However, reading vertically down Tables 5 and 6 shows that increasing the complexity of the debiased model, $d(x)$, always improves performance. The complexity, and hence learning capacity, of the debiased model is a critical factor in the overall performance of wULE. This accords with our intuition that the debiased model must learn more complex feature representations to solve the problem differently from the biased model. Overall, we see that while changing the underlying model architecture can affect performance, the wULE algorithm can function independently of any particular module architecture.

5.11 XAI Analysis: GradCAM

In this section, we use the GradCAM (Selvaraju et al., 2017) eXplainable AI (XAI) method to investigate the features that are considered most salient by the wULE debiased model, *i.e.*, the model trained to be robust to spurious correlations.

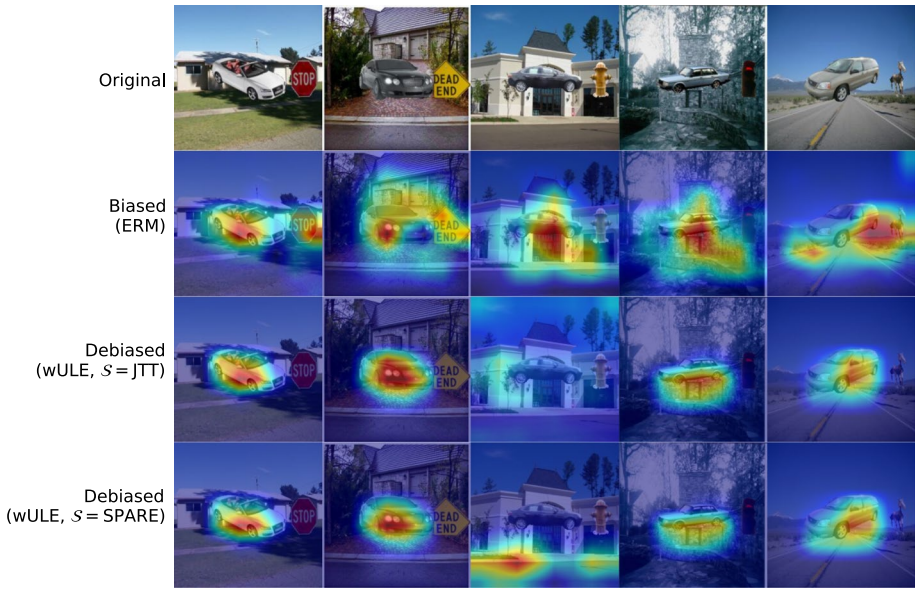


Fig. 8 GradCAM visualisations on the UrbanCars dataset for wULE debiased model trained using either the JTT or SPARE algorithms for estimating spurious correlations $\mathcal{S}$. Saliency maps for the wULE biased model trained using standard neural network training, *i.e.*, ERM are also visualised

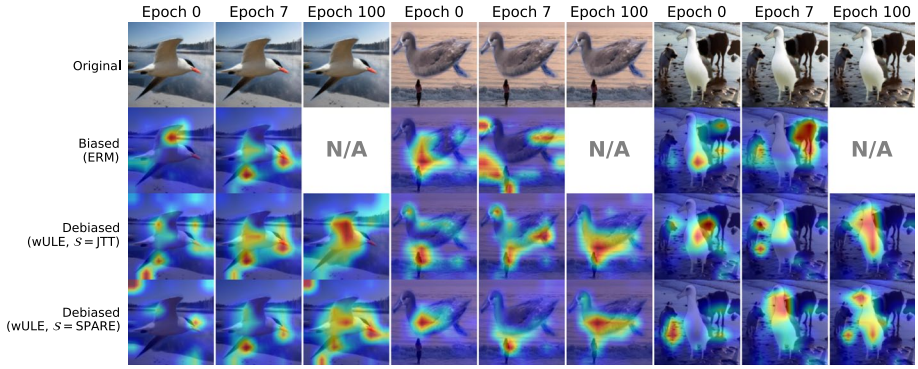


Fig. 9 GradCAM heatmaps of the biased model, $b(x)$, for ERM and the debiased model, $d(x)$, for wULE on examples from the Waterbirds dataset, showing clearer encapsulation of the bird objects — including a failure case in the middle example. Note — the biased (ERM) model is not updated after epoch 7

We visualise the saliency maps of two variants of the debiased model, trained using the JTT and SPARE-based methods for estimating $\mathcal{S}$. As a baseline for comparison, we compare the debiased model's saliency maps with those of the biased model. Recall, the biased model is trained using standard neural network training, *i.e.*, ERM. Hence, it is not expected to show robustness to spurious correlations. We visualise the saliency maps for the models on the Waterbirds and UrbanCars datasets, where the UrbanCars dataset includes several concurrent types of spurious correlation.

The GradCAM saliency maps for the UrbanCars dataset are shown in Figure 8. Recall — the goal in the UrbanCars dataset is to classify the category of the car given two types of spurious correlation: the background and a co-occurring object.

The results show that for the biased model, trained using standard training techniques, *i.e.*, ERM, both the co-occurring object and background are regarded as highly salient in most cases. In contrast, both the JTT and SPARE variants of the wULE-trained model focus on the salient object, the car, while giving little saliency to the background and co-occurring object. This demonstrates the robustness of the wULE method to multiple types of spurious correlation. It shows that with wULE training, the debiased model is capable of learning from the mistakes of the ERM-trained biased model to increase its focus on the most relevant part of the input. A failure case is shown in the middle column. In this case, the ERM-trained model does not place much saliency on the co-occurring object, which may explain why the wULE model does not function correctly in this example. Overall these results demonstrate the effectiveness of wULE for increasing robustness to multiple different types of spurious correlations.

For the Waterbirds dataset, we visualise the GradCAM heatmaps at different epochs during training. The goal is to understand how different models assign saliency to image features over the training process. Recall that the biased and debiased models are initially trained in parallel. At a specified epoch, k, the biased model training is paused while the debiased model's training continues. We, therefore, visualise the GradCAM saliency maps at epoch 0, before training, epoch $k = 7$, which is a near-optimal epoch for estimating $\mathcal{S}$, and epoch 100, where training is stable.

Figure 9 shows that at epoch $k = 7$, the biased and debiased models are already diverging. The ERM-trained biased model often focuses more on the background, while the wULE

Table 7 Results of proposed wULE with spurious estimation variants (JTT and SPARE) and comparison to state-of-the-art methods on the Waterbirds, CelebA, UrbanCars and MultiNLI datasets. All methods use a ResNet-50 architecture

	Group-Label Required in Training?	**Waterbirds** (1 Spurious × 2 Classes)		**CelebA** (1 Spurious × 2 Classes)		**UrbanCars** (2 Spurious × 2 Classes)		**MultiNLI** (1 Spurious × 3 Classes)	
		Worst-Group	Average	**Worst-Group**	Average	**Worst-Group**	Average	**Worst-Group**	Average
ERM Vapnik (1991)	✗	$62.6_{\pm 0.3}$	$97.3_{\pm 1.0}$	$47.7_{\pm 2.1}$	$94.9_{\pm 0.3}$	28.4	97.6	67.9	82.4
ERM on Class-balanced dataset	✗	$62.8_{\pm 1.6}$	$97.1_{\pm 0.1}$	$46.1_{\pm 1.5}$	$95.2_{\pm 0.4}$	33.7	98.1	*N/A*	*N/A*
George Sohoni et al. (2020)	✗	$76.2_{\pm 2.0}$	$95.7_{\pm 0.5}$	$54.9_{\pm 1.9}$	$94.6_{\pm 0.2}$	35.2	97.9	*N/A*	*N/A*
PGI Ahmed et al. (2021)	✗	$79.5_{\pm 1.9}$	$95.5_{\pm 0.8}$	$85.3_{\pm 0.3}$	$87.3_{\pm 0.1}$	34.0	95.7	*N/A*	*N/A*
EIIL Creager et al. (2021)	✗	$83.5_{\pm 2.8}$	$94.2_{\pm 1.3}$	$81.7_{\pm 0.8}$	$85.7_{\pm 0.1}$	50.6	95.5	61.2	79.4
LfF Nam et al. (2020)	✗	$78.0_{N/A}$	$91.2_{N/A}$	$77.2_{N/A}$	$85.1_{N/A}$	34.0	97.2	70.2	78.6
JTT Liu et al. (2021)	✗	$83.1_{\pm 3.5}$	$90.6_{\pm 0.3}$	$81.5_{\pm 1.7}$	$88.1_{\pm 0.3}$	55.8	95.9	72.6	78.6
CnC Zhang et al. (2022b)	✗	$88.5_{\pm 0.3}$	$90.9_{\pm 0.1}$	$88.8_{\pm 0.9}$	$89.9_{\pm 0.5}$	*N/A*	*N/A*	*N/A*	*N/A*
SPARE Yang et al. (2024)	✗	$91.6_{\pm 0.8}$	$96.2_{\pm 0.6}$	$\mathbf{90.3_{\pm 0.3}}$	$\mathbf{91.1_{\pm 0.1}}$	$\mathbf{76.9_{N/A}}$	96.6	*N/A*	*N/A*
PruSC Le et al. (2024)	✗	$82.5_{\pm 2.2}$	$89.4_{\pm 2.9}$	$\mathbf{89.7_{\pm 0.3}}$	$91.0_{\pm 0.3}$	*N/A*	*N/A*	*N/A*	*N/A*
D3M Jain et al. (2024)	✗	$81.0_{N/A}$	$90.4_{N/A}$	$83.8_{N/A}$	$91.0_{N/A}$	*N/A*	*N/A*	$\mathbf{75.5_{N/A}}$	81.5
ULE Mitchell et al. (2025)	✗	$89.0_{\pm 0.2}$	$89.6_{\pm 0.2}$	$85.3_{\pm 0.5}$	$87.6_{\pm 0.3}$	71.6	98.2	70.4	80.8
DFR Kirichenko et al. (2023)	✓	$90.4_{\pm 1.5}$	$94.1_{\pm 0.5}$	$80.1_{\pm 1.1}$	$89.7_{\pm 0.4}$	44.5	89.7	74.7	82.1
ERM on Group-balanced dataset	✓	$86.3_{\pm 0.3}$	$93.0_{\pm 1.5}$	$85.0_{\pm 1.1}$	$92.7_{\pm 0.1}$	73.9	92.2	*N/A*	*N/A*
GDRO Sagawa et al. (2020)	✓	$89.9_{\pm 0.6}$	$92.0_{\pm 0.6}$	$88.9_{\pm 1.3}$	$93.9_{\pm 0.1}$	75.2	91.6	77.7	81.4
wULE (Ours)	✗	$\mathbf{89.8_{\pm 0.6}}$	$90.9_{\pm 0.3}$	$88.0_{\pm 0.3}$	$88.4_{\pm 0.3}$	$\mathbf{76.2_{\pm 0.6}}$	96.6	$\mathbf{75.0_{\pm 0.5}}$	81.2

loss has encouraged the debiased models to correctly focus more on the bird. At epoch 100, both variants of wULE correctly focus on the bird and have learned to mostly ignore the background. These results demonstrate how the wULE loss allows the debiased model to learn over time from the mistakes of the ERM-trained biased model.

5.12 Overall Results and SOTA Comparison

Table 7 compares the performance of our proposed wULE approach on the Waterbirds, CelebA, UrbanCars and MultiNLI datasets with state-of-the-art methods from the literature. All methods under comparison used ResNet-50 as their backbone. For all like-for-like comparisons, we **bold** the best result. We report the results for wULE using the SPARE-based method for estimating $\mathcal{S}$, as this variant showed the best performance in our earlier experiments. The results show that wULE is in the top two methods for Waterbirds and UrbanCars, and the top four for CelebA. wULE also has a smaller standard deviation than most other methods. Across all three datasets, our wULE algorithm outperforms the published ULE results (Mitchell et al., 2025) in terms of WGA. This demonstrates the utility of our proposed changes to the ULE cost function.

We note that wULE tends to have similar average accuracy and WGA, whereas other methods tend to show higher average accuracy than WGA. However, we argue average accuracy can be misleading, whereas WGA better reflects a model's susceptibility to spurious correlations. For instance, Le et al. (2024) states *"A significant drop between average accuracy and worst-group accuracy is a strong indicator of relying on spurious correlations"*, which can result in worse performance on subgroups where the spurious correlations do not hold. Hence, we argue that the closer the average and worst-group accuracies are, the lower the likelihood that the model is relying on spurious correlations. Under this consideration, wULE provides the best-balanced performance.

6 Conclusion

In this paper, we propose Weighted UnLearning from Experience (wULE), a novel approach for mitigating spurious correlations in deep learning models. We build on and improve the ULE method by modifying the cost function so that the biased and debiased models are encouraged to diverge only for training examples likely to contain a spurious correlation.

We show that wULE performs effectively across both vision and language domains, achieving competitive results on the image-based Waterbirds, CelebA and UrbanCars datasets and the text-based MultiNLI dataset, highlighting its ability to generalise across different data types and domains. Our wULE method achieves second place on Waterbirds, UrbanCars and MultiNLI, and fourth place on the CelebA dataset, when using the same deep learning architecture as state-of-the-art methods.

We have also performed extensive experiments to study how the two models and their hyperparameters and loss functions interact during training. We study different methods for estimating the set of training examples containing spurious correlations, showing our SPARE-derived method leads to the best wULE performance. We have shown that wULE can be applied independently of model architecture, although using larger models, such as transformers, leads to improved performance over results reported in the state-of-the-art.

Finally, we use explainable AI to demonstrate how the wULE loss encourages the debiased model to focus on the correct salient objects across various datasets, even with multiple different types of spurious correlation.

The results highlight the effectiveness of wULE in enhancing model robustness to spurious correlations in various datasets and settings, making it a valuable addition to the existing methods in the literature.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10994-026-07164-1.

Author Contributions J.M: Conceptualization, Methodology, Software, Writing - Original Draft. J. MdR: Conceptualization, Methodology, Writing - Review & Editing. N.M: Conceptualization, Methodology, Writing - Original Draft, Writing - Review & Editing.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Ahmed, F., Bengio, Y., van Seijen, H., & Courville, A.C. (2021). Systematic generalisation with group invariant predictions. *9th international conference on learning representations, ICLR 2021, virtual event, austria, may 3-7, 2021.* OpenReview.net.

Arjovsky, M., Bottou, L., Gulrajani, I., & Lopez-Paz, D. (2019). Invariant risk minimization. *arXiv preprint arXiv:1907.02893*.

Chen, Y., Huang, W., Zhou, K., Bian, Y., Han, B., & Cheng, J. (2023). Understanding and improving feature learning for out-of-distribution generalization. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in neural information processing systems 36: Annual conference on neural information processing systems 2023, NeurIPS 2023, new orleans, LA, USA, december 10 - 16, 2023.*

Chen, Y., Zhou, K., Bian, Y., Xie, B., Wu, B., Zhang, Y., & Cheng, J. (2023). Pareto invariant risk minimization: Towards mitigating the optimization dilemma in out-of-distribution generalization. *The eleventh international conference on learning representations, ICLR 2023, kigali, rwanda, may 1-5, 2023.* OpenReview.net.

Creager, E., Jacobsen, J., & Zemel, R.S. (2021). Environment inference for invariant learning. In M. Meila and T. Zhang (Eds.), *Proceedings of the 38th international conference on machine learning, ICML 2021, 18-24 july 2021, virtual event* (Vol. 139, pp. 2189–2200). PMLR.

Darbinyan, R., Harutyunyan, H., Markosyan, A. H., & Khachatrian, H. (2023). Identifying and disentangling spurious features in pretrained image representations. *arXiv preprint arXiv:2306.12673.*

Deng, J., Dong, W., Socher, R., Li, L., Li, K., & Fei-Fei, L. (2009). Imagenet: A large-scale hierarchical image database. *2009 IEEE computer society conference on computer vision and pattern recognition (CVPR 2009), 20-25 june 2009, miami, florida, USA* (pp. 248–255). IEEE Computer Society.

Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran, T. Solorio (eds.), *Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics: Human language technologies, NAACL-HLT 2019, minneapolis, MN, USA, june 2-7, 2019, volume 1 (long and short papers)* (pp. 4171–4186). Association for Computational Linguistics.

Duchi, J. C., Glynn, P. W., & Namkoong, H. (2021). Statistics of robust optimization: A generalized empirical likelihood approach. *Mathematics of Operations Research, 46*(3), 946–969. https://doi.org/10.1287/MOOR.2020.1085

Ghaznavi, M., Asadollahzadeh, H., Noohdani, F. H., Tabar, S. V., Hasani, H., Alvanagh, T. A., & Baghshah, M. S. (2024). Trained models tell us how to make them robust to spurious correlation without group annotation. *arXiv preprint arXiv:2410.05345*.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *2016 IEEE conference on computer vision and pattern recognition, CVPR 2016, las vegas, NV, USA, june 27-30, 2016* (pp. 770–778). IEEE Computer Society.

Izmailov, P., Kirichenko, P., Gruver, N., & Wilson, A.G. (2022). On feature learning in the presence of spurious correlations. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, A. Oh (Eds.), *Advances in neural information processing systems 35: Annual conference on neural information processing systems 2022, NeurIPS 2022, new orleans, LA, USA, november 28 - december 9, 2022.*

Jain, S., Hamidieh, K., Georgiev, K., Ilyas, A., Ghassemi, M., & Madry, A. (2024). Improving subgroup robustness via data selection. In: A. Globersons et al. (Eds.), *Advances in neural information processing systems 37: Annual conference on neural information processing systems 2024, NeurIPS 2024, vancouver, BC, canada, december 10 - 15, 2024.*

Joshi, S., Yang, Y., Xue, Y., Yang, W., & Mirzasoleiman, B. (2023). Towards mitigating spurious correlations in the wild: A benchmark & a more realistic dataset. arXiv preprint arXiv:2306.11957.

Kingma, D.P., & Ba, J. (2015). Adam: A method for stochastic optimization. Y. Bengio, Y. LeCun (Eds.), *3rd international conference on learning representations, ICLR 2015, san diego, CA, USA, may 7-9, 2015, conference track proceedings.*

Kirichenko, P., Izmailov, P., & Wilson, A.G. (2023). Last layer re-training is sufficient for robustness to spurious correlations. *The eleventh international conference on learning representations, ICLR 2023, kigali, rwanda, may 1-5, 2023.* OpenReview.net.

Le, P. Q., Schlötterer, J., & Seifert, C. (2024). Out of spuriousity: Improving robustness to spurious correlations without group annotations. *arXiv preprint arXiv:2407.14974*.

Li, Z., Evtimov, I., Gordo, A., Hazirbas, C., Hassner, T., Canton-Ferrer, C., & Ibrahim, M. (2023). A whac-a-mole dilemma: Shortcuts come in multiples where mitigating one amplifies others. *IEEE/CVF conference on computer vision and pattern recognition, CVPR 2023, vancouver, BC, canada, june 17-24, 2023* (pp. 20071–20082). IEEE.

Lin, T., Maire, M., Belongie, S.J., Hays, J., Perona, P., Ramanan, D., & Zitnick, C.L. (2014). Microsoft COCO: Common objects in context. In D.J. Fleet, T. Pajdla, B. Schiele, T. Tuytelaars (Eds.), *Computer vision - ECCV 2014 - 13th european conference, zurich, switzerland, september 6-12, 2014, proceedings, part V* (Vol. 8693, pp. 740–755). Springer.

Liu, E.Z., Haghgoo, B., Chen, A.S., Raghunathan, A., Koh, P.W., Sagawa, S., & Finn, C. (2021). Just train twice: Improving group robustness without training group information. In M. Meila, T. Zhang (Eds.), *Proceedings of the 38th international conference on machine learning, ICML 2021, 18-24 july 2021, virtual event* (Vol. 139, pp. 6781–6792). PMLR.

Liu, Z., Luo, P., Wang, X., & Tang, X. (2015). Deep learning face attributes in the wild. *2015 IEEE international conference on computer vision, ICCV 2015, santiago, chile, december 7-13, 2015* (pp. 3730–3738). IEEE Computer Society.

Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *7th international conference on learning representations, ICLR 2019, new orleans, LA, USA, may 6-9, 2019.* OpenReview.net.

Mitchell, J., del Rincón, J. M., & McLaughlin, N. (2025). Unlearning from experience to avoid spurious correlations. *Pattern Analysis and Applications, 28*(4), 195. https://doi.org/10.1007/S10044-025-01570-2

Nam, J.H., Cha, H., Ahn, S., Lee, J., & Shin, J. (2020). Learning from failure: Training debiased classifier from biased classifier. *Advances in neural information processing systems 33: Annual conference on neural information processing systems 2020, NeurIPS 2020, december 6-12, 2020, virtual* (pp. 20673–20684).

Park, S.M., Georgiev, K., Ilyas, A., Leclerc, G., & Madry, A. (2023). TRAK: attributing model behavior at scale. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett (Eds.), *International conference on machine learning, ICML 2023, 23-29 july 2023, honolulu, hawaii, USA* (Vol. 202, pp. 27074–27113). PMLR.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., & Chintala, S. (2019). Pytorch: An imperative style, high-performance deep learning library. In H.M. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E.B. Fox, R. Garnett (Eds.), *Advances in neural information processing systems 32: Annual conference on neural information processing systems 2019, NeurIPS 2019, december 8-14, 2019, vancouver, BC, canada* (pp. 8024–8035).

Sagawa, S., Koh, P.W., Hashimoto, T.B., & Liang, P. (2020). Distributionally robust neural networks. *8th international conference on learning representations, ICLR 2020, addis ababa, ethiopia, april 26-30, 2020.* OpenReview.net.

Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *IEEE international conference on computer vision, ICCV 2017, venice, italy, october 22-29, 2017* (pp. 618–626). IEEE Computer Society.

Shah, H., Tamuly, K., Raghunathan, A., Jain, P., & Netrapalli, P. (2020). The pitfalls of simplicity bias in neural networks. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin (Eds.), *Advances in neural information processing systems 33: Annual conference on neural information processing systems 2020, NeurIPS 2020, december 6-12, 2020, virtual.*

Sohoni, N.S., Dunnmon, J., Angus, G., Gu, A., & Ré, C. (2020). No subclass left behind: Fine-grained robustness in coarse-grained classification problems. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin (Eds.), *Advances in neural information processing systems 33: Annual conference on neural information processing systems 2020, NeurIPS 2020, december 6-12, 2020, virtual.*

Tiwari, R., & Shenoy, P. (2023). Overcoming simplicity bias in deep networks using a feature sieve. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett (Eds.), *International conference on machine learning, ICML 2023, 23-29 july 2023, honolulu, hawaii, USA* (Vol. 202, pp. 34330–34343). PMLR.

Vapnik, V. (1991). Principles of risk minimization for learning theory. In J.E. Moody, S.J. Hanson, R. Lippmann (Eds.), *Advances in neural information processing systems 4: Annual conference on neural information processing systems 1991, NeurIPS 1991, december 2-5, 1991, denver, colorado, USA* (pp. 831–838). Morgan Kaufmann.

Wah, C., Branson, S., Welinder, P., Perona, P., & Belongie, S. (2011). The caltech-UCSD birds-200-2011 dataset. Technical report CNS-TR-2011-001, california institute of technology.

Wang, Q., Zhou, J.P., Zhou, Z., Shin, S., Han, B., & Weinberger, K.Q. (2025). Rethinking LLM unlearning objectives: A gradient perspective and go beyond. *The thirteenth international conference on learning representations, ICLR 2025, singapore, april 24-28, 2025.* OpenReview.net.

Williams, A., Nangia, N., & Bowman, S.R. (2018). A broad-coverage challenge corpus for sentence understanding through inference. In M.A. Walker, H. Ji, A. Stent (Eds.), *Proceedings of the 2018 conference of the north american chapter of the association for computational linguistics: Human language technologies, NAACL-HLT 2018, new orleans, louisiana, USA, june 1-6, 2018, volume 1 (long papers)* (pp. 1112–1122). Association for Computational Linguistics.

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., & Rush, A.M. (2020). Transformers: State-of-the-art natural language processing. In Q. Liu, D. Schlangen (Eds.), *Proceedings of the 2020 conference on empirical methods in natural language processing: System demonstrations, EMNLP 2020 - demos, online, november 16-20, 2020* (pp. 38–45). Association for Computational Linguistics.

Yang, P., Wang, Q., Huang, Z., Liu, T., Zhang, C., & Han, B. (2025). Exploring criteria of loss reweighting to enhance LLM unlearning. In A. Singh et al. (Eds.), *Forty-second international conference on machine learning, ICML 2025, vancouver, BC, canada, july 13-19, 2025* (Vol. 267). PMLR / OpenReview.net.

Yang, Y., Gan, E., Dziugaite, G.K., & Mirzasoleiman, B. (2024). Identifying spurious biases early in training through the lens of simplicity bias. In S. Dasgupta, S. Mandt, Y. Li (Eds.), *International conference on artificial intelligence and statistics, 2-4 may 2024, palau de congressos, valencia, spain* (Vol. 238, pp. 2953–2961). PMLR.

Zhang, J., Lopez-Paz, D., & Bottou, L. (2022). Rich feature construction for the optimization-generalization dilemma. In K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, S. Sabato (Eds.), *International conference on machine learning, ICML 2022, 17-23 july 2022, baltimore, maryland, USA* (Vol. 162, pp. 26397–26411). PMLR.

Zhang, M., Sohoni, N.S., Zhang, H.R., Finn, C., & Ré, C. (2022). Correct-n-contrast: a contrastive approach for improving robustness to spurious correlations. In K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, S. Sabato (Eds.), *International conference on machine learning, ICML 2022, 17-23 july 2022, baltimore, maryland, USA* (Vol. 162, pp. 26484–26516). PMLR.

Zhang, Z., & Sabuncu, M.R. (2018). Generalized cross entropy loss for training deep neural networks with noisy labels. In S. Bengio, H.M. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, R. Garnett (Eds.), *Advances in neural information processing systems 31: Annual conference on neural information processing systems 2018, NeurIPS 2018, december 3-8, 2018, montréal, canada* (pp. 8792–8802).

Zhou, B., Lapedriza, À., Khosla, A., Oliva, A., & Torralba, A. (2018). Places: A 10 million image database for scene recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 40*(6), 1452–1464. https://doi.org/10.1109/TPAMI.2017.2723009

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.